Investigating Vanishing Gradients and Early-Layer Trainability Across Network Depth

in Deep MLPs on BloodMNIST

Victoria Tran (Uyen Tran)

October 2026

Abstract

Vanishing gradients can hinder the optimization of deep neural networks by weakening parameter gradients in early layers. This study investigates depth-dependent gradient attenuation and early-layer trainability in multilayer perceptrons (MLPs) trained on BloodMNIST. Stage A compared Sigmoid networks with 2, 5, 10, and 20 hidden layers, while Stage B compared Sigmoid, Tanh, and ReLU at a fixed depth of five hidden layers. All models used Xavier Uniform initialization and stochastic gradient descent with a learning rate of 0.1. Across three random seeds, six model configurations yielded 18 unique main training runs, each trained for 20 epochs. Measurements included layer-wise root-mean-square weight gradients, normalized first-layer weight displacement, training-loss progression, and validation performance. In Stage A, the mean first-to-last hidden-layer gradient ratio decreased from 0.4070 at D2 to approximately 2.71×10^{-12} at D20, accompanied by smaller first-layer weight displacement and slower optimization. At fixed depth, Tanh and ReLU exhibited substantially less relative gradient attenuation than Sigmoid, while Tanh achieved the highest mean validation accuracy (86.21%) and macro-F1 (0.8441). These findings demonstrate depth- and activation-dependent differences under the evaluated conditions, without establishing a unique causal mechanism or generalizability beyond the tested configurations.

1 Introduction

1.1 Problem and Motivation

Gradient-based optimization of deep neural networks depends on the propagation of loss gradients through successive layers. During backpropagation, the chain rule introduces repeated transformations involving weight matrices and activation-function derivatives. Under certain conditions, these transformations can substantially attenuate gradients reaching layers near the input, a phenomenon known as *vanishing gradients*. Small gradients with respect to early-layer parameters can result in correspondingly small updates under standard stochastic gradient descent (SGD), potentially limiting learning in those layers.

Gradient-propagation difficulties have been investigated from several theoretical and empirical perspectives. Bengio et al. (1994) examined the difficulty of learning long-term dependencies in recurrent neural networks, where gradients must propagate through repeated transformations across time. Related optimization challenges arise across the depth of feedforward networks. Glorot and Bengio (2010) investigated the effects of activation functions, gradient behavior, and weight initialization on deep feedforward-network training. Schoenholz et al. (2017) subsequently

developed a mean-field framework connecting signal propagation and network trainability under assumptions about randomly initialized networks.

These findings motivate examining gradient behavior alongside parameter movement and optimization progress, rather than relying exclusively on final classification performance. This study investigates these relationships in multilayer perceptrons (MLPs) of varying depths using BloodMNIST, a standardized biomedical image classification dataset from MedMNIST v2 (Yang et al., 2023). BloodMNIST serves as a controlled experimental testbed; the objective is to characterize depth-dependent gradient attenuation and early-layer trainability rather than optimize clinical image-classification performance.

1.2 Research Questions

The investigation addresses two research questions.

Primary Research Question (RQ1): How does increasing network depth affect layer-wise weight-gradient magnitudes and the trainability of early layers in Sigmoid MLPs on BloodMNIST?

Secondary Research Question (RQ2): Under a common initialization and training protocol, how do Sigmoid, Tanh, and ReLU differ in relative gradient attenuation, optimization progress, and classification performance at a fixed network depth?

To address RQ1, Stage A compares MLPs with 2, 5, 10, and 20 hidden layers using Sigmoid activation. To address RQ2, Stage B compares Sigmoid, Tanh, and ReLU at a fixed depth of five hidden layers. Both stages use Xavier Uniform initialization and a common SGD training protocol.

1.3 Research Hypotheses and Contributions

Three hypotheses guide the experimental investigation.

H1 — Depth Effect. Increasing network depth is expected to produce stronger attenuation of early-layer RMS weight gradients relative to those in later hidden layers under conditions susceptible to vanishing gradients.

H2 — Early-Layer Trainability Effect. Conditions exhibiting stronger early-layer gradient attenuation are expected to show smaller normalized first-layer weight displacement and slower optimization progress, as assessed through training-loss trajectories and the number of epochs required to reach a predefined loss threshold.

H3 — Activation Effect. Under a common initialization rule, Sigmoid and Tanh were hypothesized to exhibit stronger depth-dependent early-layer gradient attenuation than ReLU, with the magnitude of these differences potentially depending on initialization. Stage B evaluates activation-dependent differences at D5 only; activation-by-depth and activation-by-initialization interactions remain untested.

The empirical analysis combines layer-wise RMS weight-gradient measurements with normalized first-layer weight displacement, training-loss progression, and validation classification metrics. Measurements are aggregated across three random seeds to characterize variation between training runs. The resulting comparisons examine the relationship between network depth, gradient attenuation, and early-layer trainability under controlled conditions, together with activation-dependent differences at a fixed depth.

2 Background and Related Work

2.1 Backpropagation and Gradient Attenuation

Backpropagation computes gradients of a loss function with respect to neural-network parameters by applying the chain rule through successive layers. In a feedforward multilayer perceptron (MLP), the computation at hidden layer ℓ is

$$z^{(\ell)} = W^{(\ell)} a^{(\ell-1)} + b^{(\ell)}, \qquad a^{(\ell)} = \phi(z^{(\ell)}), \tag{1}$$

where $W^{(\ell)}$ and $b^{(\ell)}$ denote the weight matrix and bias vector, $a^{(\ell-1)}$ is the input activation, and ϕ is the hidden-layer activation function.

For a hidden layer followed by another affine transformation, the backpropagated gradient with respect to its pre-activation satisfies

$$\delta^{(\ell)} = \left(W^{(\ell+1)}\right)^{\top} \delta^{(\ell+1)} \odot \phi'(z^{(\ell)}), \tag{2}$$

where $\delta^{(\ell)} = \partial \mathcal{L} / \partial z^{(\ell)}$, $\mathcal{L}$ is the loss function, and $\odot$ denotes element-wise multiplication. Gradients reaching earlier layers therefore depend on successive transformations involving weight matrices and activation derivatives. When these transformations are sufficiently contractive, gradient magnitudes can decrease substantially with depth. Conversely, sufficiently amplifying transformations can produce exploding gradients.

Bengio et al. (1994) analyzed related difficulties in recurrent neural networks, demonstrating why gradient-based learning can become increasingly difficult as temporal dependencies grow longer. Although recurrent networks propagate gradients across time using shared parameters, whereas feedforward MLPs propagate gradients across distinct layers, both involve repeated applications of the chain rule.

An important distinction exists between backpropagated activation gradients and gradients with respect to model weights. For an individual training example, the weight gradient at layer ℓ is

$$\frac{\partial \mathcal{L}}{\partial W^{(\ell)}} = \delta^{(\ell)} \left(a^{(\ell-1)}\right)^{\top}. \tag{3}$$

Thus, weight-gradient magnitudes depend on both the backpropagated error and the activations entering the layer. Glorot and Bengio (2010) demonstrated that weight gradients and backpropagated gradients need not exhibit identical layer-wise behavior.

Accordingly, RMS weight-gradient measurements characterize parameter-gradient magnitudes but do not directly measure the full propagation of activation gradients or the singular-value properties of network Jacobians.

2.2 Activation Functions and Weight Initialization

Activation functions influence gradient propagation through their derivatives. Three activation functions are considered in this study: logistic Sigmoid, hyperbolic tangent (Tanh), and rectified linear unit (ReLU).

The logistic Sigmoid derivative is

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)), \qquad 0 < \sigma'(z) \leq 0.25. \tag{4}$$

Its derivative approaches zero for large positive or negative pre-activations, potentially contributing to gradient attenuation across successive layers. Glorot and Bengio (2010) further identified the positive mean of logistic Sigmoid activations and saturation-related learning dynamics as factors contributing to optimization difficulties in deep feedforward networks.

Tanh is zero-centered, with derivative

$$\tanh'(z) = 1 - \tanh^2(z). \tag{5}$$

Its derivative reaches 1 at the origin but approaches zero in saturated regions. Consequently, Tanh and Sigmoid differ in their local gradient behavior despite both being bounded, saturating nonlinearities.

ReLU is defined as

$$\mathrm{ReLU}(z) = \max(0, z), \tag{6}$$

with derivative 1 for positive inputs and 0 for negative inputs. Unlike saturating activations, ReLU does not attenuate gradients through its derivative along active units. However, inactive units block the corresponding gradient paths, and gradient magnitudes remain sensitive to weight transformations and initialization. Glorot et al. (2011) demonstrated the effectiveness of rectifier networks in several supervised-learning settings, including experiments in which rectifier networks achieved competitive performance without unsupervised pre-training.

Weight initialization also affects forward and backward signal magnitudes. Glorot and Bengio (2010) proposed a normalized initialization scheme, commonly known as Xavier or Glorot initialization, intended to approximately balance signal propagation under simplifying assumptions. For Xavier initialization with unit gain, the weight variance is

$$\mathrm{Var}(W) = \frac{2}{n_{\text{in}} + n_{\text{out}}}, \tag{7}$$

where n_{in} and n_{out} denote the numbers of input and output connections, respectively.

He et al. (2015) subsequently derived an initialization scheme accounting for rectifier nonlinearities. For standard ReLU with fan-in scaling, the corresponding weight variance is

$$\mathrm{Var}(W) = \frac{2}{n_{\text{in}}}. \tag{8}$$

These initialization rules reflect different assumptions about activation-dependent signal propagation. He et al. (2015) demonstrated that rectifier-aware initialization can improve the optimization of very deep rectified networks relative to initialization schemes not specifically designed for rectifiers.

The present experiments use Xavier Uniform initialization with unit gain across all activation conditions. This common initialization rule supports controlled comparisons but does not establish how the activation functions would compare under separately optimized initialization schemes.

2.3 Previous Findings on Gradient Propagation and Trainability

Gradient attenuation has implications for optimization, but gradient magnitude alone does not fully determine whether a network can be trained successfully. Parameter initialization, activation dynamics, and the propagation of information across layers can jointly influence optimization behavior.

Glorot and Bengio (2010) investigated these relationships in deep feedforward networks, linking unfavorable gradient and activation behavior to training difficulties. Their analysis motivated initialization schemes that approximately preserve signal magnitudes across layers. Subsequent work on rectifier networks established that activation functions and initialization must be considered jointly when analyzing trainability (Glorot et al., 2011; He et al., 2015).

Schoenholz et al. (2017) developed a mean-field theory of signal propagation in randomly initialized networks. Their framework identified characteristic depth scales governing how information propagates through network layers and related different theoretical regimes to vanishing or exploding gradients. Their experimental analyses further associated signal-propagation properties with the maximum trainable depth under the investigated conditions.

These findings provide a theoretical basis for examining layer-wise gradients together with parameter displacement and optimization progress. However, the mean-field quantities analyzed by Schoenholz et al. (2017), including theoretical criticality parameters and characteristic depth scales, are distinct from the RMS weight-gradient measurements used here. The present experiments therefore assess empirical relationships between gradient attenuation and trainability rather than directly test the underlying mean-field predictions.

3 Methodology

3.1 Dataset and Preprocessing

Experiments were conducted using BloodMNIST, an eight-class biomedical image classification dataset included in MedMNIST v2 (Yang et al., 2023). BloodMNIST comprises microscopic peripheral blood cell images derived from the dataset introduced by Acevedo et al. (2020). Each image contains three RGB channels at a resolution of 28×28 pixels.

The official dataset partitions were retained without modification, comprising 11,959 training images, 1,712 validation images, and 3,421 test images. No additional data splitting or augmentation was performed.

Images were converted to floating-point tensors with pixel intensities scaled to $[0, 1]$. Channel-wise means and standard deviations were computed exclusively from the training split. Each RGB channel was subsequently normalized according to

$$\tilde{x}_c = \frac{x_c - \mu_c}{\sigma_c}, \tag{9}$$

where x_c denotes the scaled pixel intensity in channel c, and μ_c and σ_c represent the corresponding training-set mean and standard deviation. The same normalization statistics were applied to all three partitions, preventing validation- or test-set information from entering the preprocessing estimates.

Training data were shuffled and processed in mini-batches of 128 images. Validation and test DataLoaders used batch sizes of 1,024 without shuffling. The test split was loaded for dataset

verification but was not used for model selection or classification-performance reporting.

3.2 MLP Architecture

A configurable multilayer perceptron (MLP) was implemented in PyTorch to investigate network-depth effects within a constant-width architecture.

Each normalized image was flattened into a 2,352-dimensional input vector. The network consisted of D fully connected hidden layers, each containing 256 units, followed by a linear output layer producing eight class logits. The architecture was defined as

$$2352 \rightarrow \underbrace{256 \rightarrow \cdots \rightarrow 256}_{D \text{ hidden layers}} \rightarrow 8, \tag{10}$$

where D denotes the number of hidden layers, excluding the output layer.

The same activation function was applied after every hidden layer within a given model. The output layer produced unnormalized logits, which were passed directly to the cross-entropy loss function.

No convolutional layers, batch normalization, dropout, or residual connections were included. Hidden-layer width remained fixed across all configurations, although the total number of trainable parameters increased with network depth.

3.3 Experimental Design: Stages A and B

The experimental design comprised two stages. Stage A investigated the effect of increasing network depth under Sigmoid activation. Stage B compared activation functions at a fixed depth selected after the Stage A analysis.

Stage A — Network Depth Comparison. Four MLP architectures were evaluated: D2, D5, D10, and D20, corresponding to 2, 5, 10, and 20 hidden layers, respectively. All configurations used Sigmoid activation and shared the same initialization rule, optimizer, learning rate, batch size, training duration, preprocessing procedure, and random-seed set.

Network depth was the only intentionally varied experimental factor. The analysis examined layer-wise RMS weight-gradient magnitudes, relative early-layer gradient attenuation, normalized first-layer weight displacement, training-loss progression, and validation performance.

Stage B — Activation Function Comparison. D5 was selected following the Stage A analysis because it exhibited substantial early-layer gradient attenuation while retaining measurable optimization progress. Sigmoid, Tanh, and ReLU were subsequently compared using this fixed architecture.

The Stage B configurations used the same hidden-layer width, Xavier Uniform initialization rule, SGD optimizer, learning rate, training duration, and preprocessing procedure. The same random seeds were used across activation conditions to align initial parameter values and training-data shuffling sequences for corresponding runs. Only the hidden-layer activation function was intentionally varied.

Both stages used three random seeds: 0, 1, and 2. Stage A comprised 12 training runs across four depths and three seeds. Stage B contained nine activation–seed configurations, including three D5–Sigmoid runs reused from Stage A. Consequently, the experiment comprised 18 unique main training runs, excluding preliminary verification and pilot experiments.

The design permits controlled comparisons of depth under Sigmoid activation and activation functions at D5. It does not evaluate activation-by-depth interactions.

3.4 Training Protocol and Reproducibility

All models were trained using multiclass cross-entropy loss and standard stochastic gradient descent (SGD). Every linear-layer weight matrix, including the output layer, was initialized using Xavier Uniform initialization with gain 1 (Glorot and Bengio, 2010). All biases were initialized to zero.

SGD used a fixed learning rate of 0.1, with zero momentum and zero weight decay. No adaptive optimizer or learning-rate scheduler was employed.

A preliminary learning-rate sanity pilot was conducted using the shallowest (D2) and deepest (D20) Sigmoid architectures with seed 0. Each model was trained for three complete epochs at the provisional learning rate of 0.1. The pilot was used to assess numerical stability and inspect training-loss behavior across the two depth extremes. Neither configuration exhibited non-finite losses or obvious optimization instability; therefore, the learning rate was retained without an additional search.

A training-loss threshold of 1.50 was selected from the pilot trajectories as an operational measure of optimization progress. Both the learning rate and threshold were fixed before the main experiments.

Each main run consisted of 20 complete training epochs. For every mini-batch, previously accumulated gradients were cleared, model logits and cross-entropy loss were computed, and backpropagation was performed. Layer-wise gradient measurements were collected before the corresponding SGD parameter update. After all training batches in an epoch, the updated model was evaluated on the complete validation split.

Randomness was controlled by setting Python, NumPy, and PyTorch random seeds. Each training run used a separately seeded PyTorch generator for DataLoader shuffling. Models and optimizers were initialized independently for each run, while the same seed set was applied across experimental conditions.

The recorded software environment included Python 3.13, PyTorch 2.14.1, and MedMNIST 3.0.2. Computation was performed using the Apple Silicon Metal Performance Shaders (MPS) backend. The implementation also supported CPU execution when MPS was unavailable.

Run-level records included experimental configurations, epoch-wise training and validation metrics, layer-wise gradient measurements, and first-layer weight displacement. Consistency checks verified training-history completeness, gradient-recording coverage, finite numerical outputs, and correct reuse of the D5–Sigmoid baseline.

3.5 Gradient and Trainability Measurements

Layer-wise RMS weight gradients, relative gradient magnitudes, and normalized first-layer weight displacement were measured to characterize gradient attenuation and early-layer parameter movement. Training-loss trajectories and a predefined loss threshold were used to assess optimization progress.

Layer-wise RMS weight-gradient magnitude. For hidden layer ℓ, let $G^{(\ell)}$ denote the gradient tensor of the mini-batch loss with respect to the layer's weight matrix. Its root-mean-square magnitude was defined as

$$g_\ell = \mathrm{RMS}(G^{(\ell)}) = \sqrt{\frac{1}{N_\ell}\sum_{i=1}^{N_\ell}\left(G_i^{(\ell)}\right)^2}, \tag{11}$$

where N_ℓ is the number of elements in the weight-gradient tensor. Bias gradients and output-layer gradients were excluded from the layer-wise analysis.

RMS magnitude normalizes the Euclidean gradient norm by the square root of the tensor's element count, reducing dependence on parameter-matrix size. However, it does not eliminate differences arising from layer dimensions or input-activation distributions.

To improve numerical stability when handling very small gradients, the implementation computed the Euclidean norm after rescaling the tensor by its maximum absolute element, then restored the original scale.

Gradients were recorded immediately after backpropagation and before SGD parameter updates. The recording protocol captured measurements at every training step during the first 100 global steps and from the first 10 mini-batches of every epoch.

For each hidden layer and epoch, the sampled batch-level RMS values were averaged:

$$\bar{g}_{\ell,e} = \frac{1}{10}\sum_{b=1}^{10} g_{\ell,e,b}, \tag{12}$$

where e denotes the epoch and b indexes the sampled mini-batches. The same recording protocol was applied across experimental conditions.

Relative early-layer gradient magnitude. Relative gradient attenuation was characterized using the ratio between the first and last hidden layers:

$$R_e = \frac{\bar{g}_{1,e}}{\bar{g}_{D,e}}, \tag{13}$$

where $\bar{g}_{1,e}$ and $\bar{g}_{D,e}$ denote the epoch-level mean RMS weight gradients of the first and last hidden layers, respectively.

Smaller values indicate weaker first-layer weight gradients relative to the last hidden layer. Epoch-1 ratios were used for the principal comparisons across network depths and activation functions. Ratios were calculated separately for each seed before statistical aggregation.

This metric quantifies relative weight-gradient magnitudes rather than the proportion of an activation-gradient signal preserved through backpropagation. Weight gradients depend on both backpropagated error signals and the activations entering each layer.

Normalized first-layer weight displacement. Early-layer parameter movement was quantified by the normalized distance between the current and initial weight matrices of the first hidden layer:

$$C_1(e) = \frac{\|W_1(e) - W_1(0)\|_F}{\|W_1(0)\|_F}, \tag{14}$$

where $W_1(0)$ denotes the initial weight matrix, $W_1(e)$ denotes its value after epoch e, and $\|\cdot\|_F$ is the Frobenius norm.

The initial weight matrix was preserved before training. Displacement was recorded after each epoch, with the epoch-20 value used for final comparisons.

This metric measures net parameter displacement from initialization rather than the cumulative magnitude of SGD updates. Successive parameter updates may partially cancel, producing a net displacement smaller than their accumulated magnitudes.

Training-loss threshold. Optimization progress was additionally assessed using the first epoch at which the mean training cross-entropy loss reached a predefined threshold of 1.50:

$$T_{1.50} = \min \{e \in \{1, \ldots, 20\} : \mathcal{L}_{\text{train}}(e) \leq 1.50\}, \tag{15}$$

The epoch-level training loss was computed as the sample-weighted mean of mini-batch losses collected during training. Because model parameters were updated between mini-batches, this quantity represents the average loss observed throughout an epoch rather than the loss of a single fixed model evaluated over the entire training split.

Runs that did not reach the threshold within 20 epochs were recorded as *Not reached*. The threshold-reaching epoch provides a fixed-budget indicator of optimization progress, not a criterion for mathematical convergence.

3.6 Evaluation Metrics and Statistical Reporting

Training loss and validation metrics were recorded after each epoch. Training loss was aggregated from mini-batch cross-entropy values weighted by batch size. Validation metrics were computed on the complete validation split after each epoch, without gradient computation or parameter updates.

Validation performance was evaluated using cross-entropy loss, classification accuracy, and macro-averaged F1 score. Accuracy was defined as the proportion of correctly classified validation images. Macro-F1 was computed as the unweighted arithmetic mean of the eight class-specific F1 scores, giving equal weight to each class regardless of its frequency. Undefined class-specific F1 scores were assigned zero.

For each experimental condition, run-level metrics were aggregated across the three random seeds and reported as mean ± sample standard deviation. Sample standard deviation was computed using $n - 1$ in the denominator, where $n = 3$ for metrics available from all runs.

For the training-loss threshold, the number of seeds reaching the threshold was reported separately. Mean and sample standard deviation of threshold-reaching epochs were calculated only among runs that satisfied the criterion. Conditions with no successful runs were reported as *Not reached*, and standard deviation was undefined when fewer than two runs reached the threshold.

Final classification metrics refer to validation performance after epoch 20. No held-out test-set classification results or formal statistical significance tests were reported.

4 Experimental Results

Results are reported for the two experimental stages described in Section 3. Stage A examines network depth under a fixed Sigmoid–Xavier–SGD configuration, whereas Stage B compares activation functions at a fixed depth of five hidden layers. Unless otherwise specified, reported values are means ± sample standard deviations across three random seeds.

4.1 Stage A — Depth-Dependent Gradient Attenuation

Figure 1 presents the layer-wise RMS weight-gradient magnitudes measured during epoch 1 for the D2, D5, D10, and D20 networks. Panel (a) shows absolute gradient magnitudes, while Panel (b) shows gradients normalized by the last hidden layer within each depth–seed run.

A pronounced decrease in early-layer gradient magnitude was observed with increasing depth. The mean first-layer RMS gradient decreased from $6.259 \times 10^{-4} \pm 4.655 \times 10^{-5}$ in D2 to $3.664 \times 10^{-15} \pm 5.195 \times 10^{-16}$ in D20, a difference of approximately 11 orders of magnitude. The intermediate architectures exhibited progressively smaller first-layer gradients, with mean values of 8.823×10^{-6} for D5 and 6.227×10^{-9} for D10.

The first-to-last hidden-layer RMS gradient ratios followed the same pattern, decreasing from 0.4070 ± 0.0448 in D2 to 0.006369 ± 0.001129 in D5, $4.837 \times 10^{-6} \pm 6.791 \times 10^{-7}$ in D10, and $2.711 \times 10^{-12} \pm 1.922 \times 10^{-13}$ in D20. The logarithmic gradient profiles showed substantially smaller gradients in layers closer to the input, particularly in the deeper networks.

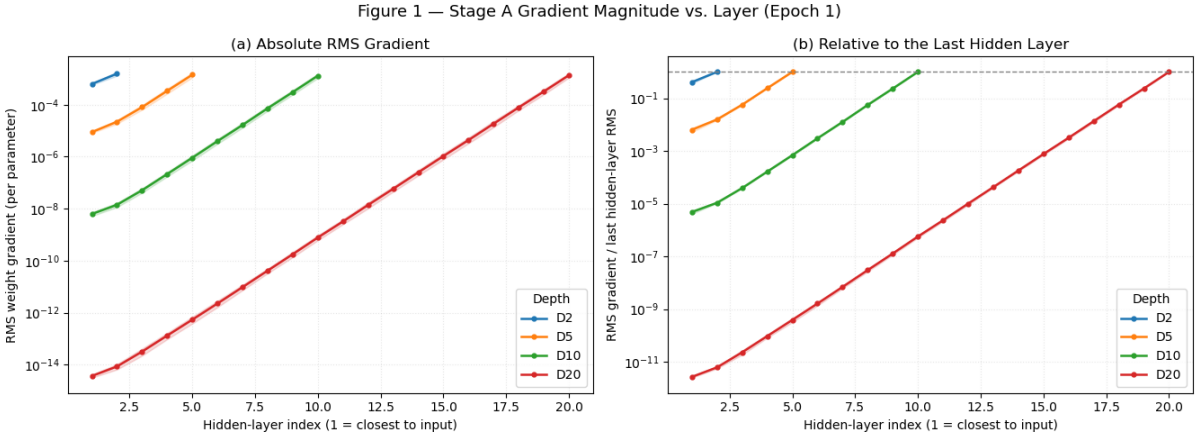


Figure 1: Layer-wise RMS weight-gradient magnitudes at epoch 1 for Sigmoid MLPs with 2, 5, 10, and 20 hidden layers. (a) Absolute RMS weight-gradient magnitude by hidden-layer index. (b) Relative RMS gradient normalized by the last hidden layer within each depth–seed run. Measurements were averaged over the first 10 training batches of epoch 1. Curves show the mean across three random seeds, with shaded regions representing ±1 sample standard deviation where supported by the logarithmic scale. Both panels use logarithmic vertical axes. Hidden Layer 1 is closest to the input.

Figure 2 extends the analysis to the complete 20-epoch training period for D20. The heatmap displays the base-10 logarithm of the mean RMS weight-gradient magnitude across three seeds for each hidden layer and epoch.

Severe early-layer gradient attenuation persisted throughout training. The mean RMS gradient of Hidden Layer 1 was 3.664×10^{-15} at epoch 1, 3.236×10^{-15} at epoch 10, and 2.918×10^{-15} at epoch 20. In contrast, the corresponding gradients of Hidden Layer 20 were 1.362×10^{-3}, 9.118×10^{-4}, and 8.794×10^{-4}.

The heatmap showed a persistent difference of approximately 11–12 orders of magnitude between the earliest and latest hidden layers. Although gradient magnitudes varied over time, the first-layer gradients remained extremely small throughout the observed training period.

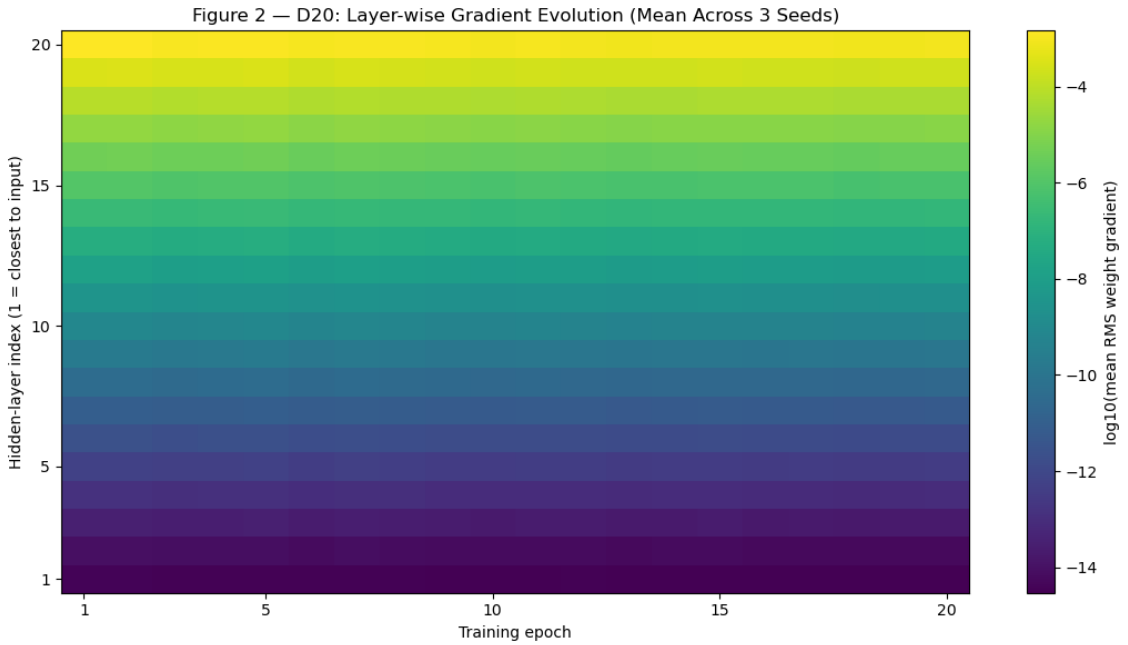


Figure 2: Evolution of layer-wise RMS weight-gradient magnitudes in D20 over 20 training epochs. The heatmap displays hidden-layer index against training epoch, with color representing $\log_{10}$ of the mean RMS weight-gradient magnitude across three random seeds. Each layer–epoch measurement is based on the first 10 training batches of that epoch. Hidden Layer 1 is closest to the input. The color scale represents the logarithm of the mean RMS gradient, not the mean of logarithmically transformed gradients.

4.2 Stage A — Early-Layer Trainability and Classification Performance

Table 1A summarizes the Stage A gradient ratios, normalized first-layer weight displacement, training-loss progression, and final validation performance.

Normalized first-layer weight displacement decreased substantially with depth. After 20 epochs, the mean displacement was 0.3279 for D2, 0.1090 for D5, and 2.740×10^{-5} for D10. All three D20 runs recorded zero first-layer displacement at the available numerical precision. This recorded value does not establish that the underlying mathematical parameter updates were identically zero.

Figure 3 shows the mean training-loss trajectories over 20 epochs. D2 exhibited rapid loss reduction, reaching a final mean training loss of 0.3963 ± 0.0065. D5 showed a substantially slower decrease, with a final mean loss of 1.2555 ± 0.0086. D10 and D20 remained near a training loss of 2.01 throughout the training period, with nearly overlapping trajectories.

All three D2 runs reached the predefined training-loss threshold at epoch 2. D5 reached the threshold between epochs 14 and 15, with a mean of 14.67 ± 0.58 epochs. Neither D10 nor D20 reached the threshold within the 20-epoch training budget.

Final validation performance followed a similar depth-dependent pattern. D2 achieved the highest mean validation accuracy (0.8123 ± 0.0112) and macro-F1 (0.7851 ± 0.0208). D5 achieved lower values of 0.4663 ± 0.0195 and 0.3377 ± 0.0213, respectively. D10 and D20 produced identical aggregated validation accuracy (0.1820 ± 0.0126) and macro-F1 (0.0385 ± 0.0022), despite differing substantially in network depth.

Table 1A: Stage A — Network Depth Experiment. Sigmoid activation; Xavier Uniform initialization; SGD (learning rate = 0.1); 20 epochs; three random seeds.

Depth	Gradient ratio (E1)	W1 displacement (E20)	Train loss (E20)
D2	4.070e-01 ± 4.482e-02	3.279e-01 ± 2.546e-03	0.3963 ± 0.0065
D5	6.369e-03 ± 1.129e-03	1.090e-01 ± 4.262e-03	1.2555 ± 0.0086
D10	4.837e-06 ± 6.791e-07	2.740e-05 ± 8.056e-06	2.0114 ± 0.0013
D20	2.711e-12 ± 1.922e-13	0.000e+00 ± 0.000e+00	2.0117 ± 0.0014

Depth	Epochs to threshold	Seeds reached	Val accuracy	Val macro-F1
D2	2.00 ± 0.00	3/3	0.8123 ± 0.0112	0.7851 ± 0.0208
D5	14.67 ± 0.58	3/3	0.4663 ± 0.0195	0.3377 ± 0.0213
D10	Not reached	0/3	0.1820 ± 0.0126	0.0385 ± 0.0022
D20	Not reached	0/3	0.1820 ± 0.0126	0.0385 ± 0.0022

Note. Values are mean ± sample standard deviation across three seeds. E1 and E20 denote epochs 1 and 20, respectively. W1 displacement is the normalized net change in first-layer weights from initialization. The predefined training-loss threshold is 1.50. "Not reached" indicates that the threshold was not attained within 20 epochs.

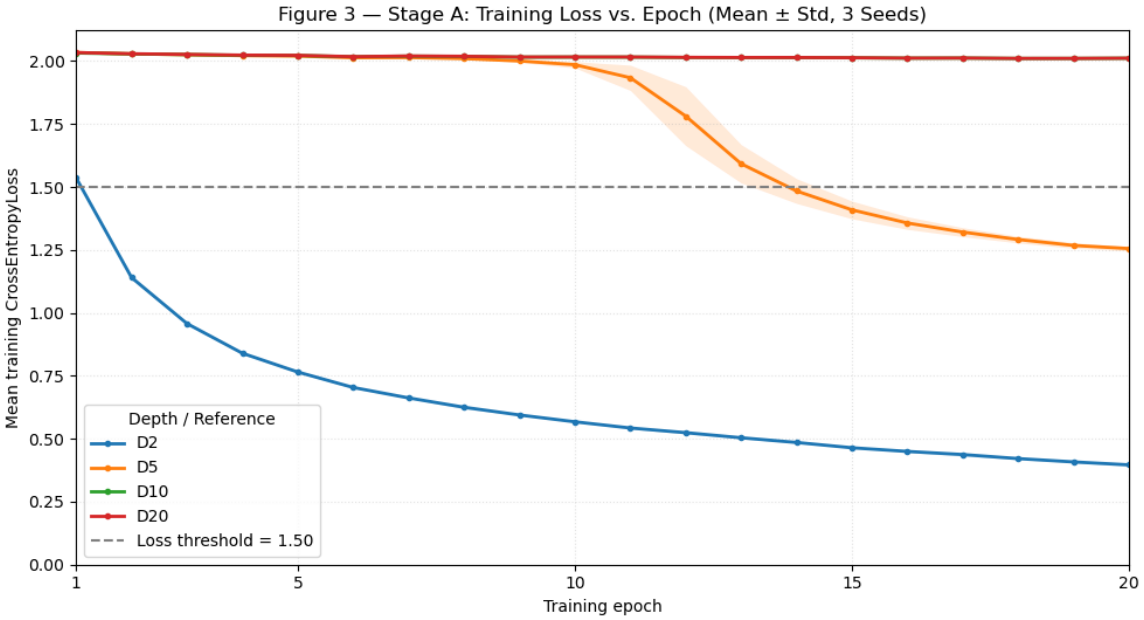


Figure 3: Training cross-entropy loss over 20 epochs for D2, D5, D10, and D20 under the fixed Sigmoid–Xavier–SGD configuration. Curves show mean training loss across three random seeds, with shaded regions representing ±1 sample standard deviation. The dashed horizontal line marks the predefined training-loss threshold of 1.50. Training losses were accumulated during mini-batch parameter updates rather than recomputed using a single fixed model at the end of each epoch.

4.3 Stage B — Activation Function Comparison

Stage B compared Sigmoid, Tanh, and ReLU in the D5 architecture under the same Xavier Uniform initialization and SGD training protocol. Table 1B presents the results across three random seeds.

Table 1B: Stage B — Activation Function Experiment. Fixed D5 architecture; Xavier Uniform initialization; SGD (learning rate = 0.1); 20 epochs; three random seeds.

Activation	Gradient ratio (E1)	W1 displacement (E20)	Train loss (E20)
Sigmoid	6.369e-03 ± 1.129e-03	1.090e-01 ± 4.262e-03	1.2555 ± 0.0086
Tanh	4.750e-01 ± 3.762e-02	3.981e-01 ± 5.189e-03	0.1934 ± 0.1173
ReLU	4.954e-01 ± 2.151e-02	4.033e-01 ± 3.480e-03	0.1992 ± 0.0765

Activation	Epochs to threshold	Seeds reached	Val accuracy	Val macro-F1
Sigmoid	14.67 ± 0.58	3/3	0.4663 ± 0.0195	0.3377 ± 0.0213
Tanh	1.00 ± 0.00	3/3	0.8621 ± 0.0079	0.8441 ± 0.0102
ReLU	1.00 ± 0.00	3/3	0.8170 ± 0.0322	0.7946 ± 0.0317

Note. Values are mean ± sample standard deviation across three seeds. E1 and E20 denote epochs 1 and 20. The predefined training-loss threshold is 1.50. The Sigmoid results are reused directly from the D5 condition in Stage A.

Activation choice was associated with substantial differences in early-layer gradient magnitudes. At epoch 1, the mean first-to-last hidden-layer RMS gradient ratio was 0.006369 for Sigmoid, compared with 0.4750 for Tanh and 0.4954 for ReLU. The corresponding absolute first-layer RMS gradient magnitudes were 8.823×10^{-6}, 3.338×10^{-3}, and 1.991×10^{-3}, respectively.

Thus, ReLU exhibited the highest mean relative gradient ratio, whereas Tanh exhibited the highest absolute first-layer RMS gradient magnitude. The differences between these two measurements reflect their distinct normalization and interpretation.

Tanh and ReLU also exhibited greater normalized first-layer weight displacement than Sigmoid. Their final mean displacement values were 0.3981 and 0.4033, respectively, compared with 0.1090 for Sigmoid.

Differences in training-loss progression were similarly pronounced. All Tanh and ReLU runs reached the predefined threshold by epoch 1, whereas Sigmoid required 14–15 epochs. At epoch 20, mean training losses were 0.1934 for Tanh, 0.1992 for ReLU, and 1.2555 for Sigmoid.

Tanh achieved the highest observed final validation performance, with mean accuracy of 0.8621 ± 0.0079 and macro-F1 of 0.8441 ± 0.0102. ReLU achieved mean accuracy of 0.8170 ± 0.0322 and macro-F1 of 0.7946 ± 0.0317, while Sigmoid achieved 0.4663 ± 0.0195 and 0.3377 ± 0.0213, respectively.

Although ReLU showed a slightly higher mean relative gradient ratio and first-layer weight displacement than Tanh, Tanh achieved higher mean validation accuracy and macro-F1 under the evaluated configuration. These differences are descriptive results from three seeds; no statistical significance tests were performed.

The Stage B results establish activation-dependent differences under a shared initialization and fixed network depth. They do not characterize how these differences vary with network depth or alternative initialization schemes.

5 Discussion

5.1 H1 — Network Depth and Gradient Attenuation

The Stage A results support H1 within the evaluated Sigmoid–Xavier–SGD configuration. Increasing network depth from two to twenty hidden layers was associated with progressively stronger attenuation of early-layer RMS weight gradients relative to those in later hidden layers. The layer-wise profiles in Figure 1 demonstrate a pronounced depth-dependent pattern, while the D20 measurements in Figure 2 indicate that severe early-layer attenuation persisted throughout the 20-epoch training period.

These observations are consistent with established accounts of gradient attenuation in deep networks. Repeated multiplication by weight matrices and activation-function derivatives can reduce the magnitude of gradients propagated toward earlier layers. Bengio et al. (1994) analyzed related difficulties across recurrent transformations over time, while Glorot and Bengio (2010) investigated the effects of activation functions and initialization on gradient behavior in deep feedforward networks. In particular, the bounded derivative of the logistic Sigmoid function provides a plausible contribution to attenuation across successive hidden layers. However, the present experiment did not directly measure activation saturation or layer-wise Jacobian properties, and therefore cannot establish the relative contributions of these mechanisms.

The findings are also qualitatively consistent with the mean-field analysis of Schoenholz et al. (2017), which relates network depth, initialization, and signal propagation to trainability. Their theoretical framework characterizes randomly initialized networks under specific statistical assumptions, whereas the present study measures RMS weight gradients in finite-width MLPs during training. The observed gradient ratios should therefore not be interpreted as estimates of theoretical criticality parameters or characteristic depth scales.

Taken together, the Stage A measurements provide empirical evidence of substantial depth-dependent weight-gradient attenuation under the tested conditions. They support H1 without establishing that increasing depth necessarily produces vanishing gradients across other activation functions, initialization schemes, or neural-network architectures.

5.2 H2 — Gradient Attenuation and Early-Layer Trainability

The results support H2 as an empirical association between early-layer gradient attenuation and reduced optimization progress. Across Stage A, models exhibiting smaller early-layer RMS weight gradients generally showed smaller normalized first-layer weight displacement and slower training-loss reduction. The deeper Sigmoid networks also failed to reach the predefined training-loss threshold within the fixed training budget, as summarized in Table 1A and illustrated by Figure 3.

Under standard SGD without momentum or weight decay, each parameter update is proportional to its corresponding loss gradient. Consequently, very small early-layer weight gradients provide a plausible explanation for limited parameter movement during training. The observed relationship between gradient magnitude and weight displacement is consistent with this optimization mechanism.

However, normalized first-layer weight displacement measures net movement from initialization rather than the cumulative magnitude of individual parameter updates. Moreover, the reported training-loss threshold represents an operational measure of optimization progress

within 20 epochs, not a mathematical convergence criterion. The zero first-layer displacement recorded for D20 must also be interpreted in the context of finite numerical precision rather than as evidence of exactly zero underlying gradients or updates.

The association between gradient attenuation and trainability is consistent with previous research. Glorot and Bengio (2010) demonstrated that unfavorable gradient and activation dynamics can accompany optimization difficulties in deep feedforward networks. Schoenholz et al. (2017) further connected signal-propagation characteristics with trainable network depth under the conditions investigated in their theoretical and empirical analyses.

Nevertheless, the present results do not establish gradient attenuation as the sole cause of weaker optimization. Increasing depth also changes the number of parameters and the composition of successive nonlinear transformations. These factors may influence optimization independently of, or jointly with, the measured weight-gradient magnitudes. The results therefore support a relationship between stronger early-layer attenuation, smaller recorded parameter displacement, and slower learning, while leaving the precise causal contributions unresolved.

5.3 H3 — Activation-Dependent Differences

The Stage B results provide evidence of activation-dependent differences in gradient attenuation and trainability at fixed D5 depth. Under the shared Xavier Uniform initialization and SGD protocol, Tanh and ReLU exhibited substantially larger early-layer RMS weight gradients relative to the last hidden layer than Sigmoid. Both activations were also associated with greater normalized first-layer weight displacement, earlier attainment of the training-loss threshold, and stronger validation performance.

These findings are consistent with the importance of activation-function properties in deep-network optimization. Glorot and Bengio (2010) identified limitations of logistic Sigmoid activations in deep feedforward networks, including saturation-related behavior. Glorot et al. (2011) demonstrated that rectifier networks could achieve competitive supervised-learning performance, motivating the use of ReLU as an alternative to saturating nonlinearities. The Stage B results are consistent with these broader findings but do not directly identify activation saturation or inactive ReLU units as the mechanisms responsible for the observed differences.

The comparison between Tanh and ReLU warrants more cautious interpretation. ReLU exhibited a slightly higher mean first-to-last hidden-layer RMS gradient ratio, whereas Tanh produced a larger absolute first-layer RMS gradient and higher mean final validation accuracy and macro-F1. Thus, the relative gradient ratio, absolute gradient magnitude, and classification performance did not yield identical rankings. Stronger relative early-layer gradients or greater net parameter displacement cannot, by themselves, establish superior classification performance.

Initialization is an additional consideration. All three activation functions were evaluated using Xavier Uniform initialization with unit gain. Although this common rule preserves a controlled comparison, it does not necessarily provide equivalent signal-propagation conditions for different nonlinearities. He et al. (2015) demonstrated the importance of rectifier-aware initialization for training very deep ReLU networks. Accordingly, the present results characterize activation differences under a shared initialization rule, rather than performance under activation-specific initialization schemes.

H3 is therefore partially supported as originally formulated. The experiment establishes substantial activation-dependent differences at D5, particularly between Sigmoid and the Tanh/ReLU

alternatives. However, the small difference in mean relative gradient ratios between Tanh and ReLU, together with the limited number of seeds and absence of formal significance testing, does not establish a reliable ranking between those two activations. Furthermore, Stage B did not vary network depth or initialization, so the proposed dependence of activation effects on these factors remains untested.

6 Limitations

Several limitations constrain the interpretation and generalizability of the experimental findings.

The experiments were conducted exclusively on BloodMNIST, using low-resolution 28×28 RGB images and the official training and validation splits. Although this dataset provides a consistent setting for controlled comparisons, the observed gradient and optimization behavior may differ on higher-resolution images, other biomedical datasets, or tasks with different input distributions and classification complexity. The official test split was not used for final performance evaluation; consequently, the reported validation metrics do not provide an independent estimate of test-set generalization or establish clinical utility.

The architectural investigation was restricted to constant-width, fully connected MLPs without normalization layers, residual connections, or other gradient-preserving architectural components. Stage A varied the number of hidden layers while maintaining a width of 256 units, but total parameter count increased with depth. The resulting differences therefore cannot be attributed to network depth independently of all changes in model capacity. Stage B compared activation functions only at D5, which was selected following the Stage A analysis. Activation-by-depth interactions were not evaluated, and the fixed-depth activation comparison should be interpreted within this selected experimental setting rather than as a comprehensive comparison across architectures.

All models used Xavier Uniform initialization with unit gain and plain SGD at a fixed learning rate of 0.1. The learning rate was retained following a limited stability pilot involving D2 and D20, rather than a systematic hyperparameter search. Although a common initialization and optimization protocol supported controlled comparisons, these settings may not be equally suitable for all architectures and activation functions. In particular, rectifier-specific initialization has been shown to affect the optimization of deep ReLU networks (He et al., 2015). The 20-epoch training budget further limits conclusions about whether poorly performing configurations might improve with longer training or alternative optimization settings.

The gradient measurements also have methodological limitations. Layer-wise RMS values were computed from hidden-layer weight gradients, with epoch-level estimates based on the first 10 training mini-batches of each epoch. These measurements characterize parameter-gradient magnitudes over a consistent sampling window but do not represent exhaustive measurements over every mini-batch. Moreover, weight-gradient RMS values depend on layer-input activations and parameter dimensions; they do not directly measure activation-gradient propagation, Jacobian singular values, or the fraction of a gradient signal preserved across layers. Activation saturation and inactive ReLU units were not measured directly. Similarly, normalized first-layer weight displacement quantifies net movement from initialization rather than the cumulative magnitude of individual SGD updates.

Finite numerical precision introduces an additional consideration when interpreting extremely small gradients and parameter changes. Although gradient norms were computed using a

rescaling procedure to improve numerical stability, this procedure does not eliminate floating-point limitations in gradient computation or parameter updates. In particular, all D20 runs recorded zero normalized first-layer weight displacement at the available numerical precision. This observation does not establish that the underlying mathematical gradients or parameter updates were identically zero; sufficiently small updates may be lost through finite-precision rounding.

Statistical uncertainty is another limitation. Each configuration was evaluated using three random seeds, and results were summarized using the mean and sample standard deviation. While this captures some variability associated with initialization and training-data shuffling, three runs provide limited information about the distribution of possible outcomes. No confidence intervals or formal statistical significance tests were conducted. Consequently, relatively small differences, particularly between Tanh and ReLU, should not be interpreted as statistically established performance rankings.

The optimization and evaluation protocol further restricts the conclusions. Epoch-level training loss was accumulated during mini-batch updates rather than recomputed using a fixed model after each epoch. The training-loss threshold of 1.50, selected after the preliminary pilot and fixed before the main experiments, provides an operational measure of optimization progress within the specified training budget rather than a formal convergence criterion. Failure to reach this threshold within 20 epochs does not establish that a model is incapable of learning under different training conditions.

Finally, the observed associations between early-layer gradient attenuation, parameter displacement, and optimization performance do not establish a unique causal mechanism. Increasing network depth changes both model capacity and the sequence of nonlinear transformations, while activation functions influence several aspects of representation learning beyond gradient magnitude. The results are consistent with gradient-attenuation-related optimization difficulties, but they do not establish that vanishing gradients alone account for the observed differences in trainability or validation performance.

7 Future Work

Several extensions could further investigate the mechanisms underlying gradient attenuation and assess the generalizability of the observed relationships between gradient behavior and trainability.

A primary direction is to examine interactions among network depth, activation functions, and weight initialization. A broader factorial design could compare Sigmoid, Tanh, and ReLU across multiple MLP depths under both Xavier and He initialization. Such experiments would extend the fixed-depth activation comparison in Stage B and help determine whether activation-dependent differences in gradient attenuation and optimization vary with network depth and initialization. In particular, comparing ReLU under Xavier and He initialization would provide a direct assessment of initialization sensitivity, motivated by the findings of He et al. (2015). Additional measurements of activation saturation, inactive ReLU units, and backpropagated activation gradients could help distinguish potential mechanisms underlying the observed weight-gradient patterns.

A second direction is to investigate architectural approaches intended to improve optimization in deep networks. Incorporating batch normalization and residual connections into the

experimental framework would allow their effects on layer-wise gradient magnitudes, first-layer parameter displacement, and training-loss progression to be evaluated. These experiments could determine whether the severe early-layer attenuation observed in deeper Sigmoid MLPs persists when alternative architectural components are introduced.

Optimization sensitivity also warrants investigation. Comparisons between standard SGD and adaptive optimizers such as Adam, together with systematic learning-rate experiments and longer training budgets, could clarify the extent to which the observed differences depend on the optimization protocol. Such comparisons would require careful interpretation because adaptive optimization changes the relationship between raw gradient magnitude and parameter-update size. Additional training runs and random seeds would also permit more reliable characterization of variability across experimental conditions.

Finally, extending the analysis beyond constant-width MLPs and BloodMNIST would provide evidence about the generalizability of the findings. Convolutional neural networks, including residual architectures, could be evaluated on BloodMNIST and additional biomedical image classification datasets. Experiments involving different input resolutions and model capacities would help assess whether the observed relationships persist across architectural and data settings. Independent test-set evaluation, combined with a larger number of random seeds, would provide a stronger basis for comparing generalization performance across configurations.

8 Conclusion

This study investigated depth-dependent gradient attenuation and early-layer trainability in multilayer perceptrons (MLPs) trained on BloodMNIST. Two controlled experimental stages examined the effects of network depth under Sigmoid activation and differences among Sigmoid, Tanh, and ReLU at a fixed depth. Layer-wise RMS weight gradients, normalized first-layer weight displacement, training-loss progression, and validation performance were evaluated across three random seeds.

For the primary research question (RQ1), increasing depth from two to twenty hidden layers under the Sigmoid–Xavier–SGD configuration was associated with substantial early-layer gradient attenuation. The mean first-to-last hidden-layer RMS gradient ratio decreased from 0.4070 in D2 to approximately 2.71×10^{-12} in D20. Stronger attenuation was accompanied by smaller first-layer weight displacement, slower training-loss reduction, and poorer validation performance. These results support H1 within the evaluated configuration and H2 as an empirical association between gradient attenuation and reduced early-layer trainability, without establishing a unique causal mechanism.

For the secondary research question (RQ2), activation functions exhibited markedly different gradient and optimization behavior at D5. Tanh and ReLU produced substantially larger relative early-layer gradients, greater first-layer weight displacement, and faster training-loss reduction than Sigmoid. Tanh achieved the highest mean validation accuracy and macro-F1, whereas ReLU exhibited a slightly higher mean relative gradient ratio. These findings partially support H3, demonstrating activation-dependent differences under a common initialization rule without establishing a general performance ranking or interactions with network depth and initialization.

The conclusions are limited to constant-width MLPs evaluated on BloodMNIST under the specified initialization, optimization, and training protocol. The results characterize relationships among weight-gradient attenuation, parameter movement, and optimization progress, but do

not establish their causal contributions or generalizability to other architectures and datasets. Further controlled experiments are required to distinguish the effects of depth, activation, initialization, and optimization on neural-network trainability.

References

Andrea Acevedo, Anna Merino, Santiago Alférez, Ángel Molina, Laura Boldú, and José Rodellar. A dataset of microscopic peripheral blood cell images for development of automatic recognition systems. *Data in Brief*, 30:105474, 2020. doi: 10.1016/j.dib.2020.105474.

Yoshua Bengio, Patrice Simard, and Paolo Frasconi. Learning long-term dependencies with gradient descent is difficult. *IEEE Transactions on Neural Networks*, 5(2):157–166, 1994. doi: 10.1109/72.279181.

Xavier Glorot and Yoshua Bengio. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, volume 9 of *Proceedings of Machine Learning Research*, pages 249–256. PMLR, 2010. URL https://proceedings.mlr.press/v9/glorot10a.html.

Xavier Glorot, Antoine Bordes, and Yoshua Bengio. Deep sparse rectifier neural networks. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pages 315–323. PMLR, 2011. URL https://proceedings.mlr.press/v15/glorot11a.html.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 1026–1034, 2015. doi: 10.1109/ICCV.2015.123.

Samuel S. Schoenholz, Justin Gilmer, Surya Ganguli, and Jascha Sohl-Dickstein. Deep information propagation. In *International Conference on Learning Representations*, 2017. URL https://arxiv.org/abs/1611.01232.

Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10:41, 2023. doi: 10.1038/s41597-022-01721-8.